Fields and Waves I

Lecture 1 – Why Study E&M?

Introduction to Fields and Waves

Alex Patterson

Electrical, Computer, and Systems Engineering Department

Rensselaer Polytechnic Institute, Troy, NY

These slides were prepared through the work of the following people:

- Kenneth A. Connor – ECSE Department, Rensselaer Polytechnic Institute, Troy, NY (Emeritus)
- J. Darryl Michael – GE Global Research Center, Niskayuna, NY
- Thomas P. Crowley – National Institute of Standards and Technology, Boulder, CO
- Sheppard J. Salon – ECSE Department, Rensselaer Polytechnic Institute, Troy, NY (Emeritus)
- Lale Ergene – ITU Informatics Institute, Istanbul, Turkey
- Jeffrey Braunstein – ECE Department, University at Albany
- James Lu - ECSE Department, Rensselaer Polytechnic Institute, Troy, NY
- James Dylan Rees - ECSE Department, Rensselaer Polytechnic Institute, Troy, NY
- Alex Patterson - ECSE Department, Rensselaer Polytechnic Institute, Troy, NY

Materials from other sources are referenced where they are used. Those listed as Ulaby are figures from Ulaby's textbook.

What is Electrical Engineering?

- My definition:

Manipulating electromagnetic energy to

"do stuff"

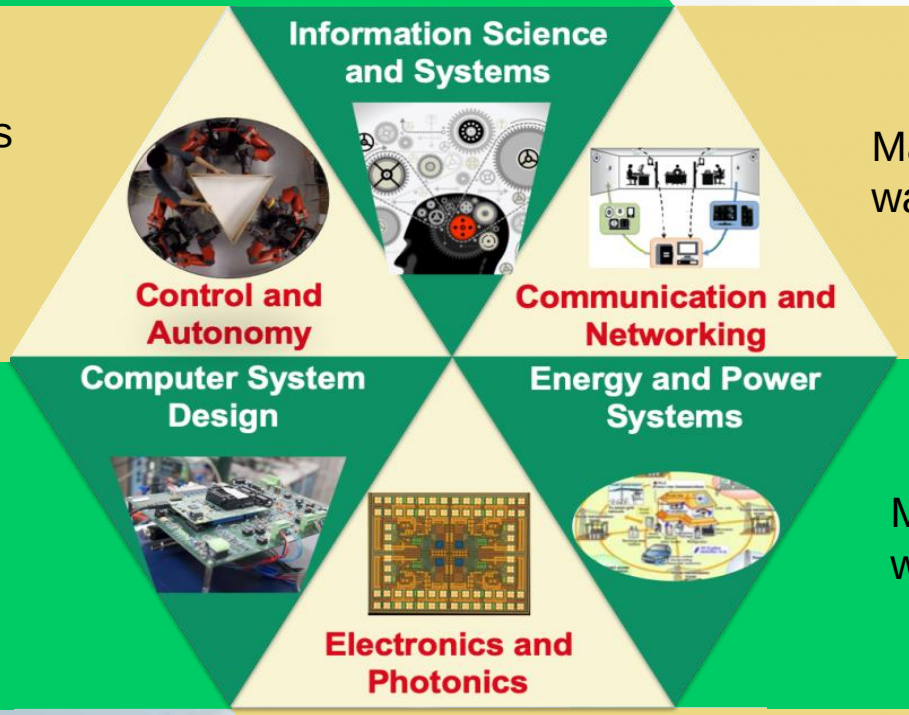
Electromagnetic energy? What kind of stuff?

What is Electrical Engineering?

Manipulating electromagnetic signals (or charge) to process data and make decisions

Using electromagnetic signals as feedback and instructions for automating systems

Manipulating electromagnetic waves to convey information



Designing electronic systems to process instructions and data (charge)

Manipulating electromagnetic waves to transmit power

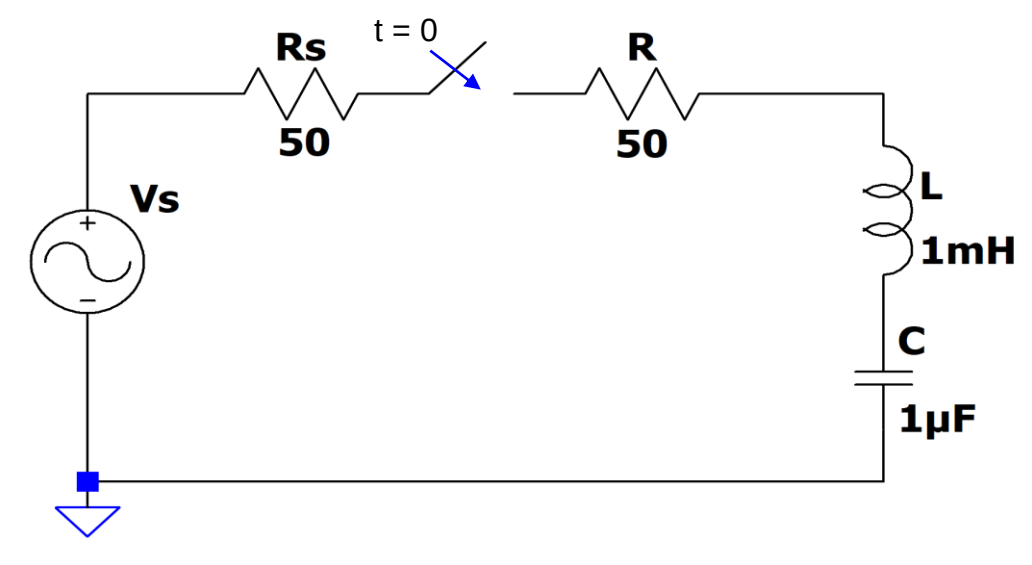
Manipulating electrons or photons to enable other technologies in hardware

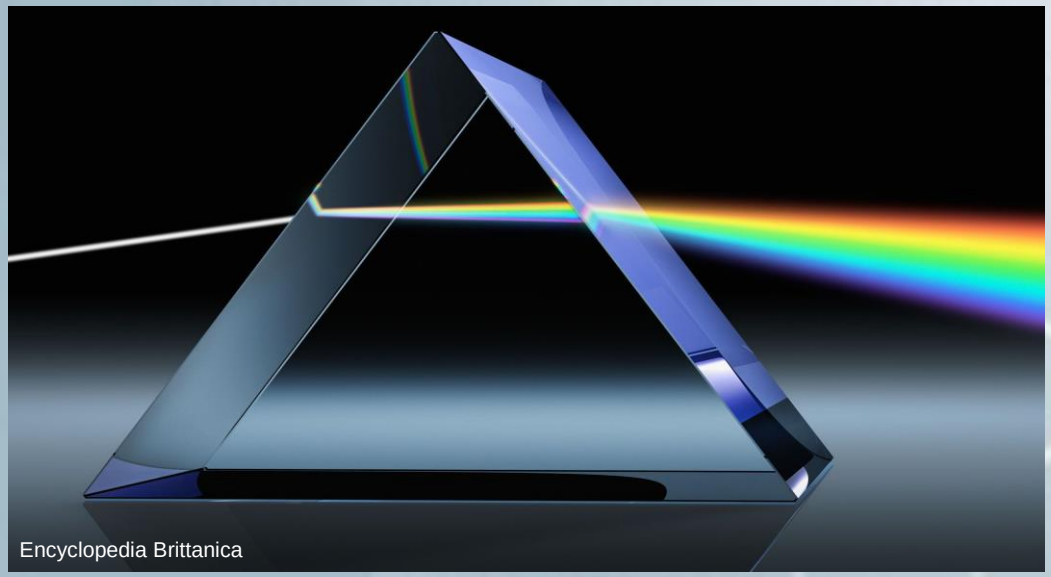
Why study electromagnetism?

But why study electromagnetism?

Can’t we just simulate it?

Why study electromagnetism?





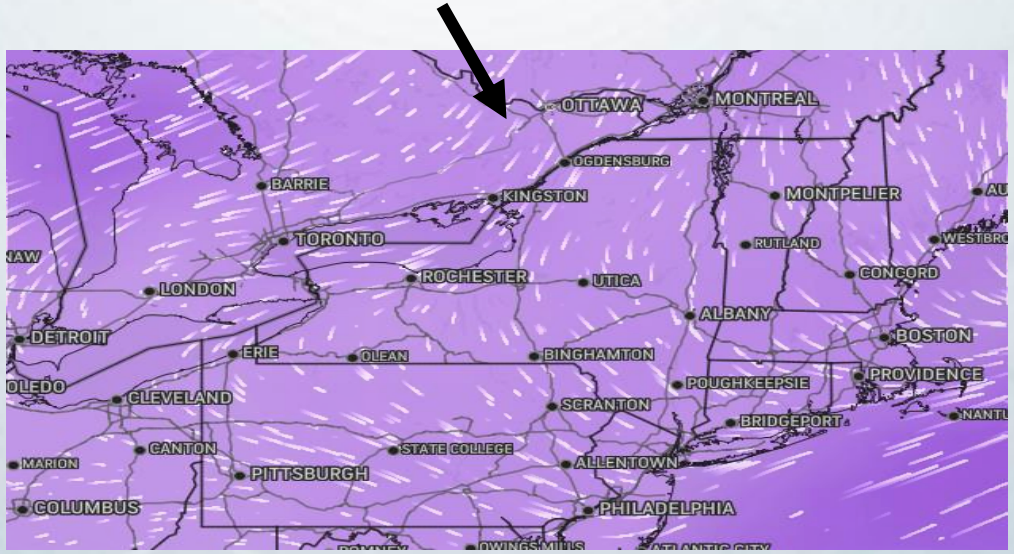
- When the switch closes at t = 0, does the voltage travel instantaneously down the wire?
- When can we assume it does?
- If not, how do we describe the voltage and current in the wire?
- How is power transmitted in circuits?
- What are resistance, capacitance, and inductance physically?
- What about the geometry and materials of R, C, and L gives them their specific numerical values?
- How do we design R, C, and L to have specific values?
- How do electromagnetic waves travel through space?
- How does the medium affect the properties of the waves?
- What happens when the wave hits the boundary between two materials?

Why use “fields” and “waves”?

- **Field**: mathematical construct representing a physical quantity – has a value at every point in space and time

Examples: temperature (scalar), wind velocity (vector)



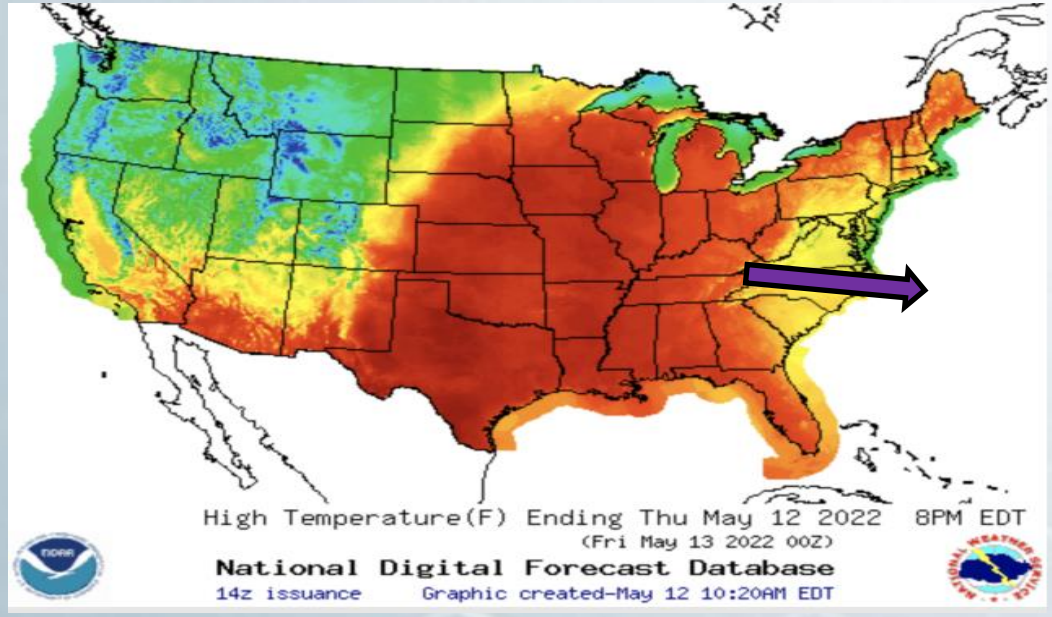


For this class:

- Scalar: voltage
- Vector: current density, electric field, magnetic field

Why use “fields” and “waves”?

- **Wave**: propagating disturbance of a physical quantity



For this class:

- Scalar: voltage, current
- Vector: electric field, magnetic field

How does this fit into our course?

Unit 1

Transmission Lines

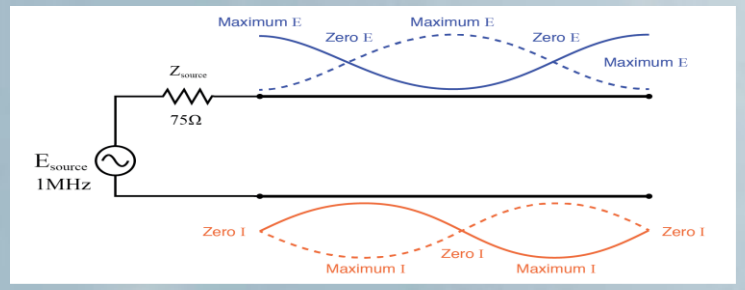
Unit 2 and Unit 3

Electrostatics & Magnetostatics

Unit 4

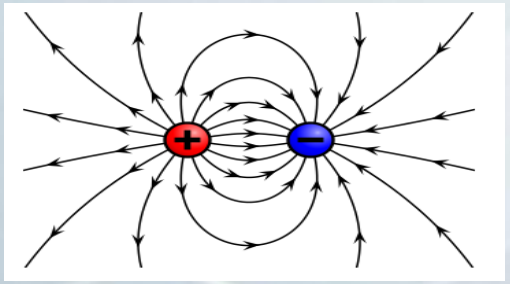
Propagating Electromagnetic Waves

Voltage & Current Waves



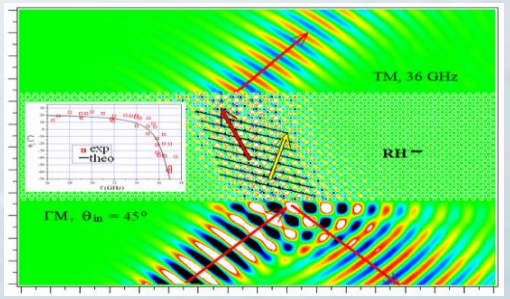
https://www.allaboutcircuits.com/uploads/articles/standing-waves-on-open-transmission-line.png

Electric & Magnetic Fields



https://brilliant.org/problems/interactions-between-charges/

Electric & Magnetic Fields & Waves



https://www.researchgate.net/figure/FDTD-simulation-of-the-36-GHz-TM1-wave-incident-at-45-o-across-the-GM-interface-of-a-2D_fig3_228926662

Why Study Electromagnetism (E&M)?

- Electromagnetism is the most **fundamental theory of electrical engineering**: the one that tells us how electric magnetic fields and related concepts (such as current and voltage) actually work and relate to each other.
- This means that by understanding E&M, you will also understand **how electrical systems work at a fundamental level**.
- You will also understand the limits of other theories such as circuit theory (i.e. where these theories break down).

Why Study Electromagnetism (E&M)?

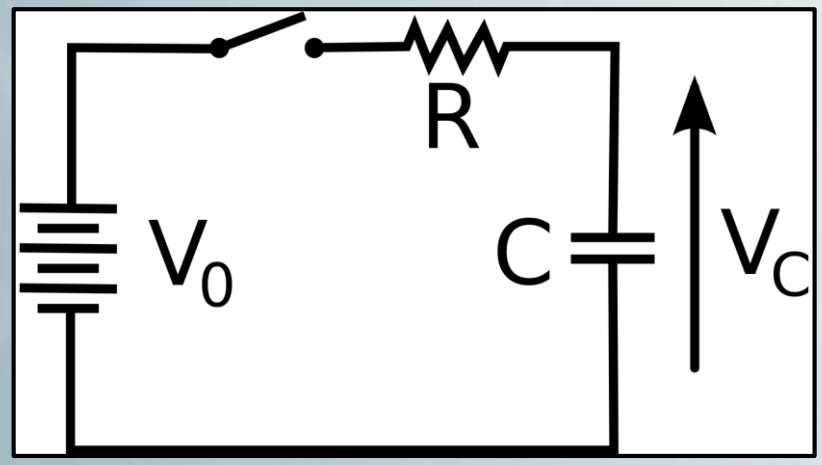
Relationship with Circuit Theory

Circuit theory uses simplified (lumped) models of components. The model, however, does not include:

- Details on how the components **actually work**
- Distributed properties (such as transmission lines)
- Electromagnetic waves (such as microwaves, radio waves, and optics)
- Noise

Why Study Electromagnetism (E&M)?

Relationship with Circuit Theory



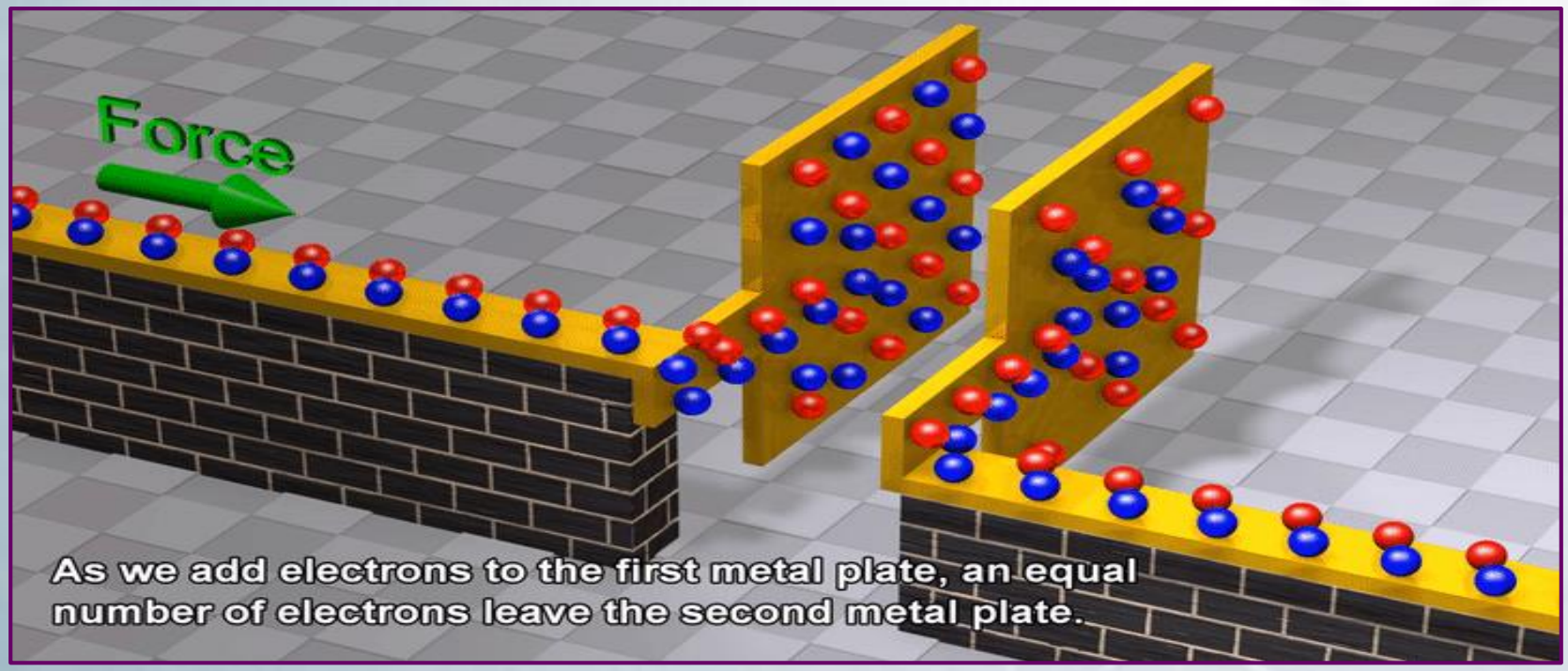
$$I(t) = \frac{V_0}{R} \cdot e^{\frac{-t}{\tau_0}}$$

$$V(t) = V_0 \left(1 - e^{\frac{-t}{\tau_0}}\right)$$

- In circuit theory, you learn equations for the current and voltage behavior of a capacitor.
- But what are the physical phenomena that *cause* a capacitor to behave this way?

Why Study Electromagnetism (E&M)?

Relationship with Circuit Theory

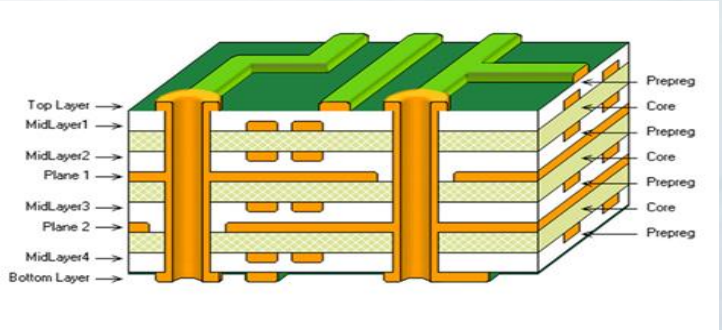


Displacement current (unit 4) is the current that flows into and out of a capacitor. It is not due to any connection between the two plates but due to the time-varying electric field between them.

Why Study Electromagnetism (E&M)?

Relationship with Circuit Theory

- Digital circuits typically process pulses having clock rates below **4 GHz**. "Via pins" provide electrical connections between the circuit planes.
- Circuit operation is *nominally* not based upon electromagnetic wave effects. However, these circuits consist of:
  - Densely packed, multiple planes of metal traces providing flow paths for the signals
  - DC power feeds
  - ground returns



https://www.circuits-diy.com/facts-about-multilayer-pcb/

- Kirchoff's current and voltage laws *fail* in most **high-speed circuits**, which must be analyzed using E&M theory, as signal power flows are not confined to the intended metal wires or circuit paths.

Crosstalk: when electromagnetic signals from one circuit or transmission line interfere with nearby circuits or transmission lines

Crosstalk occurs due to the mutual **capacitance** and **inductance** between conductors, which allows for the transfer of electromagnetic energy between them.



Why Study Electromagnetism (E&M)?

Electromagnetic wave propagation devices, such as antennas, **fundamentally cannot be described** using traditional circuit theory.

We must take into account the **geometry** of the device, the **materials** and **propagation media** involved, and - most importantly - to think of electric phenomena as being composed of **waves**.

Maxwell's Equations mathematically describe the relationship between the electric and magnetic fields, as well has how they move through media



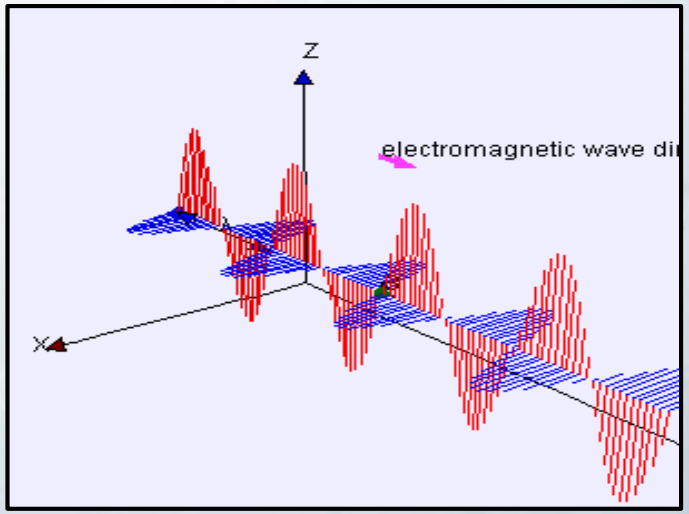
Why Study Electromagnetism (E&M)?

The "Waves" Part of Fields & Waves

Electromagnetic fields propagate as waves. They are governed by the electromagnetic wave equations, which are derived from **Maxwell's Equations** and can be written as:

$$\frac{\partial^2 \vec{E}}{\partial t^2} = c^2 \nabla^2 \vec{E} \qquad \frac{\partial^2 \vec{B}}{\partial t^2} = c^2 \nabla^2 \vec{B}$$

where **c** is the speed of light / EM waves. (Note that c depends on the medium the wave is propagating through!)



Source: https://en.wikipedia.org/wiki/Electromagnetic_radiation

Why Study Electromagnetism (E&M)?

Maxwell's Equations

$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}$	(1)	Gauss' law
$\nabla \cdot \mathbf{B} = 0$	(2)	Magnetic monopoles
$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$	(3)	Faraday's law
$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$	(4)	Ampere-Maxwell law

These four equations **fundamentally describe all of electromagnetism**.

The rest is a question of applications (i.e. this course, all of electrical engineering, etc.)

Why Study Electromagnetism (E&M)?

Maxwell's Equations

$$\nabla \cdot \mathbf{D} = \rho$$



Source: spectrumscientifics.wordpress.com

Gauss' Law: Electric charge gives rise to an electric field.

Why Study Electromagnetism (E&M)?

Maxwell's Equations

$$\nabla \cdot \mathbf{B} = 0$$

When they ask you where your magnetic monopoles are



Magnetic Monopoles: "Magnetic charges" don't exist.* Only electric charges do.

* except maybe in very esoteric branches of particle physics

Why Study Electromagnetism (E&M)?

Maxwell's Equations

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$



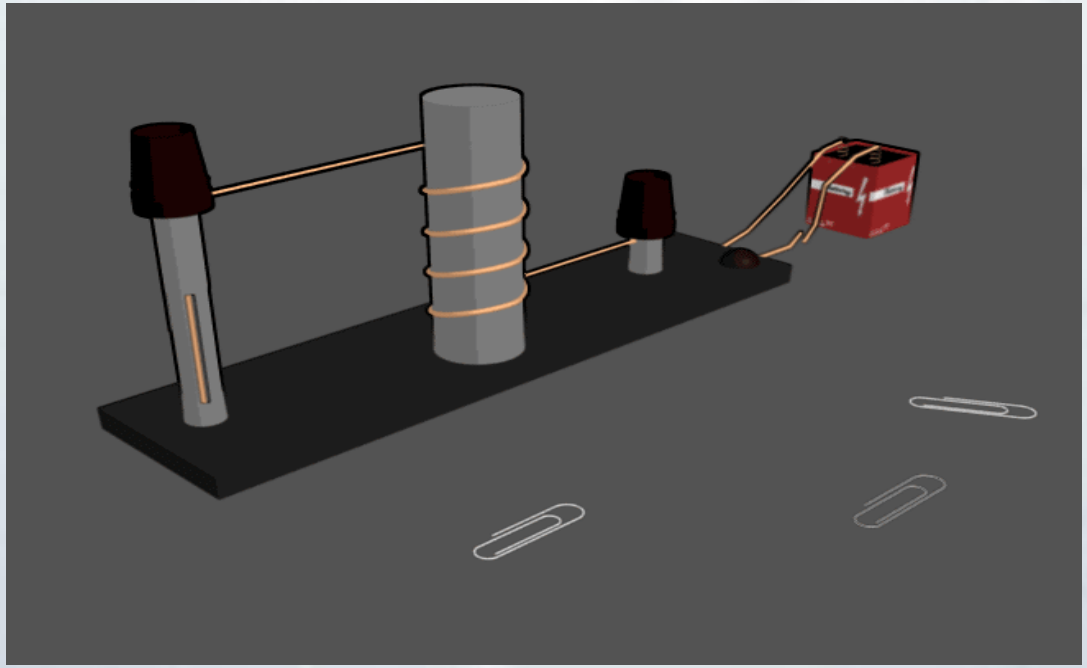
Source: Wikimedia Commons

Faraday's Law: A changing magnetic field gives rise to an electric field.

Why Study Electromagnetism (E&M)?

Maxwell's Equations

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$



Source: https://simple.wikipedia.org/wiki/Electromagnet

Ampere's Law: Current gives rise to a magnetic field.

The **propagation of electromagnetic waves** depends on the media through which they are traveling. Applying Maxwell's equations at the boundaries of different media yields **boundary conditions** which govern the behavior of the waves.

Simulation of EM waves propagating through the air and hitting a sphere with water-like permittivity



Why Study Electromagnetism (E&M)?

Back to electrical engineering. What does this all mean?

- When you have voltage (V), you have an electric field intensity (**E** in V/m)
- When you have current (I), you have a magnetic flux density (**B** in Wb/m^2).

Also:

- When you have a change in **B**, a voltage is generated
- When you have a change in **E**, a current is generated

Syllabus and Course Details

Tips and Best Practices

- **There is a lot of math in this course**: complex functions; multivariable vector calculus; different coordinate systems… Make sure you have a good basis in these.
- **Conceptual understanding is critical**. The main problem students have in this course is not the math, it's the *conceptual understanding*.

  The bridge between the problem statement and the math is the conceptual understanding – you need it to start the problem. You cannot just look at a problem in this course and "math" through it.
- **The homework is meant to prepare you for exams**. Do not take shortcuts or the exams will be a lot less fun.
- **Start the homework early**. It is time consuming to struggle with problems to learn, but it is necessary.
- **When you get stuck on a problem, stop and try to pinpoint what you don't understand**. It will help you focus your learning and help us help you in office hours.
- **Preparation for lecture**: at least skim the lecture slides and reading assignment (posted on course calendar) to know what the topics for the lecture are.

Lecture 1 Exercise

Get to know your classmates!

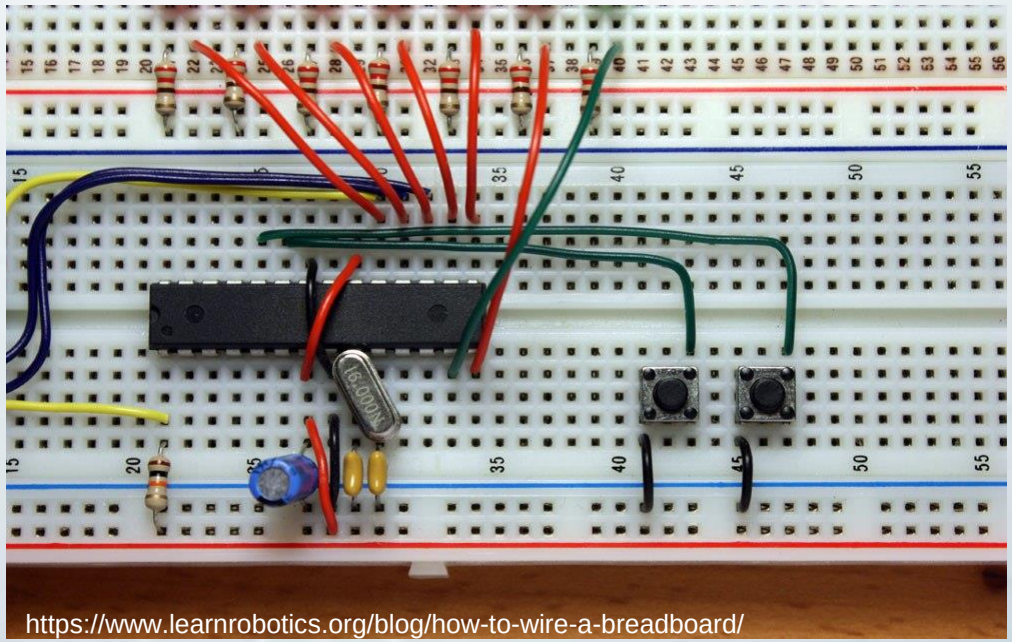
- Form a group of up to 4 and do **Lecture 01 Exercise** on Gradescope.
- Only 1 of you needs to submit, but include the names of everyone in your group.
- We'll reconvene in 10 minutes and share.

Transmission Lines

Transmission Lines

Think about circuits. What are the main assumptions we make about wires?

1. Zero **transit time** from one end to the other (speed = ∞)
2. Zero **resistance** (infinite conductivity)
3. Zero **mutual capacitance** between the wire and other conductors
4. Air has **zero conductance** (zero leakage current)
5. Zero **self-inductance** (changes in current do not generate an EMF)



https://www.learnrobotics.org/blog/how-to-wire-a-breadboard/

None of these are true, but are a good approximation for **short wires** and low **frequencies** – for everything else, we need **transmission line theory**

Transmission Line History

- Circuit theory was developed in the 18th and 19th centuries.
- One of the first practical applications was the telegraph (Francis Ronalds, 1816)
- Following this discovery, more and more **long** conductors were built for telegraphy



Transmission Line History

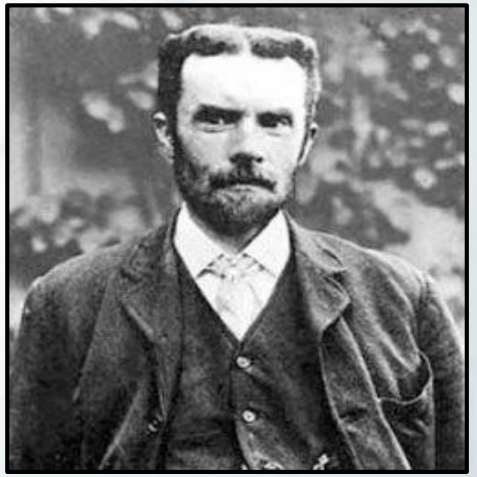
- As telegraphy expanded, there was demand for longer and longer cables
- First working undersea telegraph line laid across the English Channel in 1851, followed by other European undersea cables. The work of building a transatlantic cable began in the mid 1850s.
- These cables suffered from problems:
  - Voltage drop due to resistance
  - Signal delay
  - Signal distortion



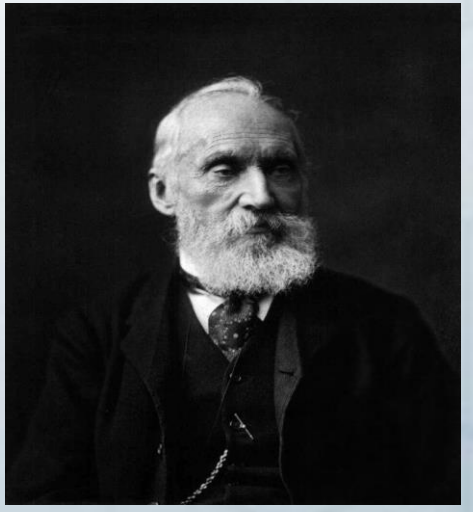
English Channel cable

Transmission Line History

- Several scientists and engineers worked to develop a **formal electrical model of long conductors**
- The equations that made up this model became known as the **Telegrapher's Equations**
- These equations allowed for intercontinental telegraph lines to develop, and they are used today for many "new" applications such as CPU microstrip lines



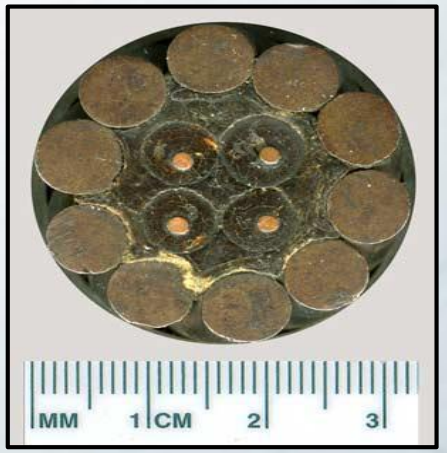
Oliver Heaviside



Lord Kelvin

Transmission Line Model

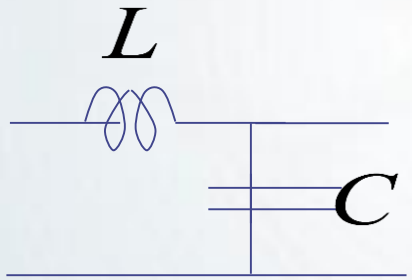
- The purpose of a transmission line is to **transmit a signal and/or power from one place to another**.
- Examples include:
  - Power Lines (60Hz)
  - Coaxial Cables
  - Twisted Pairs
  - Interconnects inside a computer



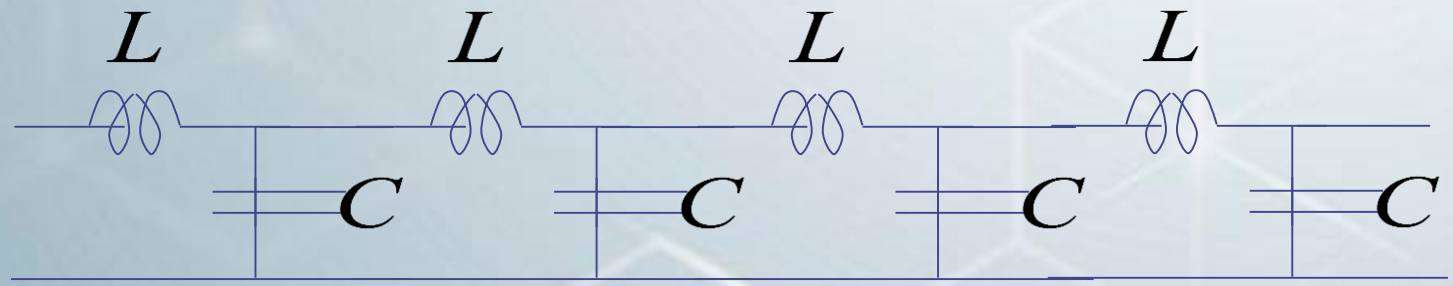
- Note that these all have **two conductors.**
- In transmission line theory, we represent a transmission line as having certain properties **per unit length**.
- Under certain conditions, we can model a short section of transmission line using circuit elements like resistors, inductors and capacitors.

Transmission Line Model

Model of a short section:



Model the full length as:



….etc.

This is where we will pick up on Monday

For Next Class

- Reading assignment to be done prior to each lecture listed on course calendar (see LMS)
- First studio session **next Wednesday** in JEC 4107. Bring your laptop and have LTspice installed!
- Homework 0 due next Wednesday at 11:59pm